# 405 HTTP Status Code in ASP.NET Core Web API

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## 405 HTTP Status Code in ASP.NET Core Web API

In this article, I will discuss how to **Return the 405 Method Not Allowed HTTP Status Code in ASP.NET Core Web API** using different approaches. Please read our previous article discussing How to Return the 404 Not Found HTTP Status Code in ASP.NET Core Web API with Examples.

### 405 HTTP Status Code

The HTTP 405 status code is called "Method Not Allowed". This error occurs when the resource does not support the HTTP request sent to the server. This means that the server recognizes the request method but does not allow it to be handled by the client API. For example, if a client sends a GET request to a URL that only accepts GET requests, the server will return a 405 error.

This is sometimes a good error code, indicating that the method was valid but not supported for the resource you are trying to access. When a service returns a 405 response, it should include an "Allow" header that lists the HTTP methods that the resource does support. For example, if a client sends a PUT request to a resource that only supports GET and POST, the server might respond with a 405 Method Not Allowed error and include the Allow: GET, POST header in the response. This helps the client understand what methods are allowed for the resource.

### Common Causes for 405 HTTP Errors

- Incorrect HTTP Method: The most common cause is using the wrong HTTP method for the endpoint. For example, you might be trying to POST to an endpoint that only supports GET.
- Unsupported HTTP Method: The server does not support the method used in the request (e.g., GET, POST, PUT, DELETE, PATCH, etc.) for the resource.
- Misconfigured Routes: In web applications, incorrect routing or route configurations in files such as Startup.cs or Program.cs.
- Server Configuration: The server or a reverse proxy might be configured to disallow certain methods for specific endpoints.

### How to Return HTTP Status Code in ASP.NET Core Web API

In ASP.NET Core Web API, we can explicitly return a HTTP status code. Method Not Allowed is using the StatusCode method, returning a HTTP 405 response. Alternatively, for configuring HTTP responses, we can use Results or IResult and return an object with the status when an HTTP method is used that isn't supported for a specific route, call or proceed and unsupported HTTP method.

#### Using HTTP Method Attributes to Return 405 Status Code

In ASP.NET Core, you can specify the allow methods with HTTP attributes. For example, we can decorate [HttpGet], [HttpPost], [HttpPut], [HttpDelete], etc. attributes on controller actions to specify the allowed methods. If a request is made using a method not specified in the attributes, the framework will automatically return a 405 error. This is a clean, declarative approach, ideal for RESTful controllers. Below, they can use the Publishers controller to demonstrate with the [HttpPut] attribute. This means we can access the endpoint using HTTP Put Request.

```
namespace Entity.Controllers.Publishers.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class PublisherController : ControllerBase
    {
        [HttpPut("{publisherId:int}")]
        [Authorize(Roles = "Admin")]
        [ProducesResponseType(StatusCodes.Status204NoContent)]
        [ProducesResponseType(StatusCodes.Status404NotFound)]
        [ProducesResponseType(StatusCodes.Status400BadRequest)]
        [ProducesResponseType(StatusCodes.Status401Unauthorized)]
        [ProducesResponseType(StatusCodes.Status403Forbidden)]
        [ProducesResponseType(StatusCodes.Status500InternalServer)]
        [ProducesResponseType(StatusCodes.Status405MethodNotAllowed)]
        [ProducesResponseType(StatusCodes.Status409Conflict)]
        [ProducesResponseType(StatusCodes.Status422UnprocessableEntity)]
        [ProducesResponseType(StatusCodes.Status429TooManyRequests)]
        [ProducesResponseType(StatusCodes.Status503ServiceUnavailable)]
        [ProducesResponseType(StatusCodes.Status504GatewayTimeout)]
        [ProducesResponseType(StatusCodes.Status505HttpVersionNotSupported)]
        [ProducesResponseType(StatusCodes.Status507InsufficientStorage)]
        [ProducesResponseType(StatusCodes.Status508LoopDetected)]
        [ProducesResponseType(StatusCodes.Status509BandwidthLimitExceeded)]
        [ProducesResponseType(StatusCodes.Status510NotExtended)]
        [ProducesResponseType(StatusCodes.Status511NetworkAuthenticationRequired)]
        [ProducesResponseType(StatusCodes.Status512ConnectionTimeout)]
        [ProducesResponseType(StatusCodes.Status513ConnectionLimitReached)]
        [ProducesResponseType(StatusCodes.Status514ConnectionRefused)]
        [ProducesResponseType(StatusCodes.Status515ConnectionReset)]
        [ProducesResponseType(StatusCodes.Status516ConnectionAborted)]
        [ProducesResponseType(StatusCodes.Status517ConnectionClosed)]
        [ProducesResponseType(StatusCodes.Status518ConnectionTimeout)]
        [ProducesResponseType(StatusCodes.Status519ConnectionLimitReached)]
        [ProducesResponseType(StatusCodes.Status520ConnectionRefused)]
        [ProducesResponseType(StatusCodes.Status521ConnectionReset)]
        [ProducesResponseType(StatusCodes.Status522ConnectionAborted)]
        [ProducesResponseType(StatusCodes.Status523ConnectionClosed)]
        [ProducesResponseType(StatusCodes.Status524ConnectionTimeout)]
        [ProducesResponseType(StatusCodes.Status525ConnectionLimitReached)]
        [ProducesResponseType(StatusCodes.Status526ConnectionRefused)]
        [ProducesResponseType(StatusCodes.Status527ConnectionReset)]
        [ProducesResponseType(StatusCodes.Status528ConnectionAborted)]
        [ProducesResponseType(StatusCodes.Status529ConnectionClosed)]
        [ProducesResponseType(StatusCodes.Status530ConnectionTimeout)]
        [ProducesResponseType(StatusCodes.Status531ConnectionLimitReached)]
        [ProducesResponseType(StatusCodes.Status532ConnectionRefused)]
        [ProducesResponseType(StatusCodes.Status533ConnectionReset)]
        [ProducesResponseType(StatusCodes.Status534ConnectionAborted)]
        [ProducesResponseType(StatusCodes.Status535ConnectionClosed)]
        [ProducesResponseType(StatusCodes.Status536ConnectionTimeout)]
        [ProducesResponseType(StatusCodes.Status537ConnectionLimitReached)]
        [ProducesResponseType(StatusCodes.Status538ConnectionRefused)]
        [ProducesResponseType(StatusCodes.Status539ConnectionReset)]
        [ProducesResponseType(StatusCodes.Status540ConnectionAborted)]
        [ProducesResponseType(StatusCodes.Status541ConnectionClosed)]
        [ProducesResponseType(StatusCodes.Status542ConnectionTimeout)]
        [ProducesResponseType(StatusCodes.Status543ConnectionLimitReached)]
        [ProducesResponseType(StatusCodes.Status544ConnectionRefused)]
        [ProducesResponseType(StatusCodes.Status545ConnectionReset)]
        [ProducesResponseType(StatusCodes.Status546ConnectionAborted)]
        [ProducesResponseType(StatusCodes.Status547ConnectionClosed)]
```